

Andrea Ninni

Born on 20/01/1998
Rome (RM), Italy

+39 3930801219
ninni.andrea98@gmail.com
andrea-ninni-21990920a
github.com/andreeedna

Skills

◦ Single-cell RNAseq analysis
◦ Bulk RNAseq analysis
◦ Genomic data analysis
◦ Linux and Bash programming
◦ R and Python programming
◦ Molecular Modeling and Dynamics
◦ Web programming ◦ Scientific Illustrations

Certificates

◦ Virtual Applied Data Science Training Institute Program (VADSTI) - Jul 2021
◦ High Performance Bioinformatics (CINECA) - Dec 2022
◦ Bioinformatic data resources Workshop (UCL) - May 2023
◦ scRNA-seq hands-on course (BITS) - Sep 2023
◦ Preparation of genomics libraries (EMBL) - Nov 2023
◦ Introduction to Deep Learning and Tensorflow (CINECA) - Apr 2024
◦ Sample Preparation for scRNA-seq experiment (Universitatsklinikum Wurzburg) - Sep 2024
◦ FELASA (Fondazione Santa Lucia) - Nov 2024

EDUCATION

2022-2025. **PhD in Evolutionary Biology and Ecology**
University of Rome "Tor Vergata"

2020-2022. **Master's Degree, MSc Bioinformatics**
University of Rome "Tor Vergata"

2016-2020. **Bachelor's Degree, BSc Biological Sciences**
University of Rome "Tor Vergata"

RESEARCH EXPERIENCE

Oct. 2021 - Jul. 2022 **Master's Thesis Student** at the University of Rome "Tor Vergata"

- Development and implementation of a bioinformatics pipeline for variant calling applied to genome sequencing data of the protected species *Caretta caretta*.
- Structural Modelling (modeling and analysis) of mitochondrial proteins.

Sep. 2022 - Oct. 2022 **Research Fellow** at IRCCS Santa Lucia
Project: *Adipose-derived vesicles: immunomodulators and biomarkers during anti-inflammatory therapy in multiple sclerosis*

- Analysis of RNAseq and scRNAseq transcriptomic data from white adipose tissue in R, in the context of multiple sclerosis.

Jun. 2024 - Dec. 2024 **Visiting Scholar** at Universitätsklinikum Würzburg

- Analysis of scRNAseq transcriptomic data from myocardium in the context of myocardial infarction.
- Experience in bone marrow cell extraction, macrophage maturation, and cell culture management.
- Experience in tissue sample processing and preparation for single-cell sequencing.

Nov. 2022 - Nov. 2025 **PhD Student** at the University of Rome "Tor Vergata"
Project: *Brown adipose tissue adaptation to environmental pressure*

- Creation and development of MACanalyzeR: an R package focused on scRNAseq analysis of the Monocyte-Macrophage cluster.
- Analysis of RNAseq and scRNAseq transcriptomic, proteomic, and metabolomic data in R and Python in the context of brown adipose tissue subjected to environmental stress.
- Handling of mice and primary cell culture.
- RNA extraction and Real-Time qPCR.

Presentations

May 2023 **UCL-Rome meeting** - Bridging the gap between experimentalists, biocurators and analysis tool providers

June 2025 **BITS 2025** - Poster: MACanalyzeR scRNAseq analysis tool reveals PPARγ^{HIGH} lipid-associated macrophages facilitate thermogenic expansion in BAT.

October 2025 **BraYn 2025** - Poster: Trem2 structural variants insight in Alzheimer's Disease and Frontotemporal Dementia.

Jan.2026 - Apr.2026 **Research Fellow** at the University of Rome "Tor Vergata"
Project: *Adipose tissue profiling by snRNAseq*

May.2026 - Ongoing **PostDoc Researcher** at the University of Rome "Tor Vergata"
Project: *Analysis of mitochondrial transfer as a novel anti-aging model*

PUBLICATIONS

Under Review - 2026 **TREM2 drives accumulation of pro-scarring monocyte-derived macrophages in the infarcted myocardium**

Under Review - 2026 **Primary Hyperparathyroidism Reveals Limited Adipose Remodeling in Human** **Co-First Author*

Under Review - 2026 **Non-canonical TERT function mitigates adipose tissue inflammation in obesity by modulating stem cell-macrophage communication and macrophage states**

Under Review - 2025 **Structural Insights into Functional Impacts of Common Variants in Mitochondrial Protein-Coding Genes of Mediterranean Loggerhead Turtles** **First Author*

Cell Death and Disease - 2026 **Cancer cachexia induces senescent reprogramming of brown adipose tissue and pro-cachectic S100A9 secretion by adipocytes**

BioPha - 2026 **Dimethyl fumarate attenuates subcutaneous adipose tissue inflammation in psoriasis**

Nat Comm - 2025 **MACanalyzeR scRNAseq analysis tool reveals PPARγ^{HIGH}/GDF15^{HIGH} lipid-associated macrophages facilitate thermogenic expansion in BAT** **First Author*

Cell Reports - 2024 **Lipid-Associated Macrophages Reshape BAT Cell Identity in Obesity** **Co-First Author*

LSA - 2024 **Frataxin deficiency shifts metabolism to promote reactive microglia via glucose catabolism**

Marine Biology - 2024 **Whole mitochondrial genome sequencing provides new insights into the phylogeography of the Mediterranean loggerhead sea turtle**

Mol Met - 2023 **Lipocalin-2 promotes adipose-macrophage interactions to shape peripheral and central inflammatory responses in experimental autoimmune encephalomyelitis**